Evanns Morales-Cuadrado

U.S. Citizen

Robotics PhD Candidate
Georgia Institute of Technology

CONTACT

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Research Website: https://www.evannsmc.com/
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TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, Matlab/Simulink
- **Frameworks & Libraries:** JAX, PyTorch, acados, CasADi, ROS 2, PX4, ArduPilot
- **Development & DevOps Tools:** Docker, Continuous Integration, Git/GitHub
- **Hardware & Experimentation:** UAV assembly, repair, and deployment
- **Motion-Capture:** OptiTrack and Vicon
- **Languages:** Spanish (Native), English (Native)

EDUCATION

Georgia Institute of Technology | 2022-2027
Ph.D. in Robotics
Doctoral Minor in Optimization Methods
- GPA: 4.0/4.0
- Advisor: Dr. Samuel Coogan
- Thesis Proposal Approved (ABD)
- Goizueta Fellow

University of Texas at Arlington | 2018-2022
Honors Bachelor of Science in Electrical Engineering
Minor in Mathematics
Minor in Physics
Certificate in Unmanned Vehicle Systems
- GPA: 3.9/4.0 Summa Cum laude
- Full National Merit Scholarship Awarded
- Electrical Engineering Honors Scholar

WORK EXPERIENCE

Graduate Researcher | AI for Autonomy — May 2026 - Present
Sandia National Laboratories — Albuquerque, NM
- Conducted applied research integrating artificial intelligence and advanced control methods for autonomous systems
- Developed and evaluated algorithms for the perception, decision-making, and control of autonomous platforms
- Implemented and tested methods in simulation and on hardware within a multidisciplinary research team

Teaching Assistant | Vertically Integrated Project in Robotics — January 2024 - Present
Georgia Institute of Technology — Atlanta, GA
- Aided professor in his instruction of this research-based course
- Provided promising undergraduates with guidance performing applied research in the field of robotics
- Taught control theoretical concepts in an accessible manner for undergraduates

Graduate Research Assistant | FACTS Lab — August 2022 - Present
Georgia Institute of Technology — Atlanta, GA
- Advised by Dr. Samuel Coogan
- Implemented novel aggressive tracking control methods on quadrotor hardware
- Developed novel nonlinear control methods for safe autonomy
- Published novel research at top conferences and journals in the fields of robotics and control theory

Research Assistant | Autonomous Systems Lab November 2021 - May 2022
University of Texas at Arlington Arlington, TX

- Advised by Dr. Frank Lewis
- Received personal mentorship from a leading researcher in the field of controls and reinforcement learning (ranked 12^{th} in the USA and 23^{rd} in the world during my time in his lab)
- Aided graduate students on reinforcement learning research applied to quadrotor control

Teaching Assistant | Graduate-Level Intelligent Systems Course January 2022 - May 2022
University of Texas at Arlington Arlington, TX

- Aided professor in his instruction of the course
- Hosted review sessions for students in preparation for exams
- Gained teaching experience in a graduate-level course while still an undergraduate

Research Assistant | Dynamical Networks and Control Lab October 2019 - May 2022
University of Texas at Arlington Arlington, TX

- Advised by Dr. Yan Wan
- Gained experience with Robot Operating System (ROS), OpenCV, and path planning
- Original research in learning-based minimum time and energy path-planning for multi-vehicle systems

SELECTED PUBLICATIONS

M. Cao, A. Harapanahalli, **E. Morales-Cuadrado**, and S. Coogan, "Trajectory Tracking for Systems with Unknown Time-Varying Disturbances." Submitted to Open Journal of Control Systems. *(under revision)*

E. Morales-Cuadrado, "Computationally Efficient and Safe Control for Aerial Robotic Systems under Threat and Disturbance." Ph.D. Thesis Proposal, Georgia Institute of Technology, 2026. *pdf available. slides available.*

E. Morales-Cuadrado, L. Chung, S. Kousik and S. Coogan, "`RTD-RAX`: Fast, Safe Trajectory Planning for Systems under Unknown Disturbances." Joint Submission to 2026 Modeling, Estimation and Control Conference (MECC) and ASME Letters in Dynamic Systems and Control (ALDSC). *(under revision) preprint available.*

E. Morales-Cuadrado, L. Baird, Y. Wardi and S. Coogan, "Lightweight Tracking Control for Computationally Constrained Aerial Systems with the Newton-Raphson Method." IEEE Transactions on Control Systems Technology, 2026. *(accepted; to be published) preprint available.*

L. Baird, **E. Morales-Cuadrado**, and S. Coogan, "Runtime Assurance for Uncertain Systems from Interval Signal Temporal Logic." Submitted to IFAC Nonlinear Analysis: Hybrid Systems (NAHS), 2025. *(under revision) preprint available.*

E. Morales-Cuadrado, C. Llanes, Y. Wardi and S. Coogan, "Newton-Raphson Flow for Aggressive Quadrotor Tracking Control." *2024 American Control Conference (ACC)*

RESEARCH INTERESTS

- Safe Autonomy
- Hardware Deployment
- Unmanned Ground and Aerial Vehicles
- Advanced Nonlinear Control
- Learning-Based Control
- Trajectory Generation and Planning

OPEN-SOURCE CONTRIBUTIONS

- **PX4 Autopilot:** Contributed documentation improvements to the official PX4 Autopilot manual.
- **Vicon4PX4:** Developed and open-sourced a ROS 2 C++ Vicon–PX4 interface enabling external vision fusion for indoor quadrotor flight experiments; adopted and showcased by Georgia Tech's Indoor Flight Lab as part of its official codebase. Also created a parallel package for OptiTrack systems.

SERVICE & LEADERSHIP

Co-Founder and President of Puerto Rican Student Association August 2024 - Present
Georgia Institute of Technology Atlanta, GA

- Addressed the need for an organization to help Puerto Rican students feel at home and stay connected to our culture
- Formed a three-member co-founder board and identified a faculty advisor
- Raised funds for the organization and recruited members from campus
- Hosted professional, social, and cultural events to meet the needs of the growing Puerto Rican student population at Georgia Tech

Undergraduate Research Mentor August 2021 - May 2022
University of Texas at Arlington Arlington, TX

- As the lead undergraduate student Dr. Frank Lewis' Autonomous Systems Lab, I mentored junior undergrads in controls research and guided them on a research project in ground vehicle control
- Guided students in Robot Operating System, Python, Linux, and Control Systems

Teaching Assistant | Graduate-Level Intelligent Systems Course January 2022 - May 2022
University of Texas at Arlington Arlington, TX

- Aided professor in his instruction of the course
- Hosted review sessions for students in preparation for exams
- Graded homeworks, exams, and projects
- Gained teaching experience in a graduate-level course while still an undergraduate

Eta Kappa Nu Public Relations Chair & Circuits I Tutor August 2020 - May 2022
University of Texas at Arlington Arlington, TX

- Executive board member of UT Arlington's IEEE-HKN chapter
- In charge of social media advertising for honor society's events and outreach initiatives for the electrical engineering student body
- Volunteered to provide free tutoring as a service to electrical engineering students in Circuits I

HONORS AND AWARDS

Top-3 Finalist at a Deep Learning Research Symposium November 2023
Professor-sponsored award for novel research in deep learning and robotics

Goizueta Fellowship at Georgia Tech August 2022
Highly selective fellowship for graduate students of Latin-American descent

Summa Cum Laude Honors at University of Texas at Arlington May 2022
Exceptional academic distinction in top GPA tier for graduation honors

Honors College Degree Requirements at University of Texas at Arlington December 2021
Completed research projects in disparate fields throughout my undergraduate career with various professors and defended a senior capstone research project before the Honors College committee

Electrical Engineering Honors Scholar at University of Texas at Arlington December 2021
Inaugural member of the electrical engineering honors cohort

IEEE-HKN International Electrical Engineering Honor Society Membership August 2021
Selective undergraduate society recognizing outstanding academic achievement and community service

Research Experience for Undergraduates Fellowship Recipient August 2020
Fellowship to sponsor promising young undergraduate researchers

Chance Vought Engineering and Science Endowment Scholarship August 2020
Highly selective award given annually to three recipients across the entire College of Engineering

Toray Composite Materials America Inc. Sons and Daughters Scholarship May 2019
Academic scholarship awarded children of employees of Toray America

James E. Casey Scholarship Recipient January 2019
Academic scholarship awarded children of employees of UPS

James R. Hoffa Memorial Scholarship Recipient May 2018
Academic scholarship awarded children of Teamsters Union Members

Grapevine, TX Rotary Club Scholarship Recipient May 2018
Local rotary club scholarship for academic and civically-minded scholars

National Hispanic Scholar March 2018
Recognition of top national performers of Latin descent on the SAT

National Merit Scholar March 2018
Recognition of top national performers on the SAT